

Type-Based Unsourced Multiple Access in Distributed MIMO: Applications to Localization and Federated Learning

Half-way seminar

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Included papers

- [A] **Kaan Okumus**, K.-H. Ngo, G. Durisi, and E. G. Ström, "Type-based unsourced multiple access over fading channels with cell-free massive MIMO," in Proc. IEEE Int. Symp. Inf. Theory (ISIT), Ann Arbor, MI, USA, June 2025.
- [B] **Kaan Okumus**, K.-H. Ngo, G. Durisi, and E. G. Ström, "Type-based unsourced multiple access over fading channels in distributed MIMO with application to multi-target localization," submitted to IEEE Trans. Wireless Commun., Mar. 2026.
- [C] **Kaan Okumus**, K.-H. Ngo, U. K. Ganesan, G. Durisi, E. G. Ström, and S. R. Pandey, "Type-based unsourced federated learning with client self-selection," accepted to IEEE Int. Conf. Commun. (ICC), Glasgow, Scotland, UK, Feb. 2026.

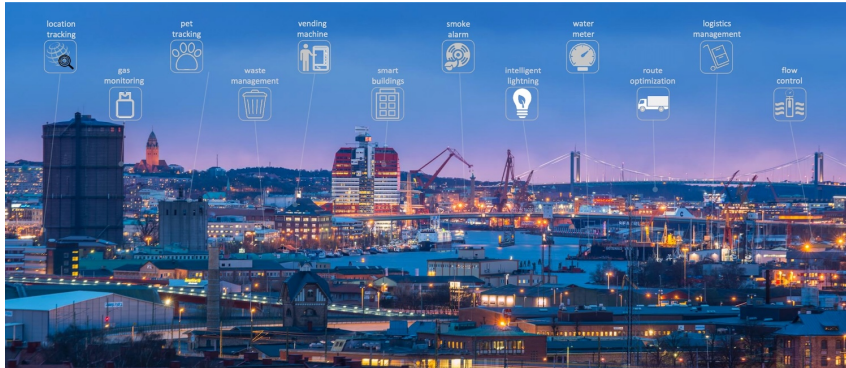
Outline

1. Introduction to type-based unsourced multiple access (TUMA)
2. TUMA over fading channels in distributed MIMO
3. Application I: Multi-target localization
4. Application II: Federated learning
5. Conclusion

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1. **Introduction to type-based unsourced multiple access (TUMA)**
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Wireless-communication-enabled massive connectivity



source: IoTpool

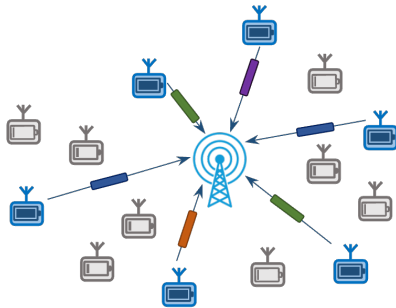
Challenge

Collect **in an energy-efficient way** data from a **massive number** of **low-cost sensors**

Massive wireless connectivity

Massive machine type communication (mMTC)

- Mostly uplink
- **Small** information payload (~ 100 bits)
- **Uncoordinated** and **sporadic** ($\leq 1/\text{min}$) transmissions
- **Massive** number of devices ($\sim 10^7/\text{Km}^2$)



Any scheme that assigns dedicated resources (time, frequency, code, etc.) to each user is not scalable

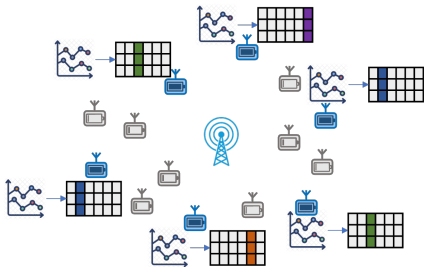
Unsourced multiple access (UMA) [Polyanskiy, 2017]

- All users employ the **same codebook/encoder**
- Receiver produces an **unordered** list of messages (no sender identification)

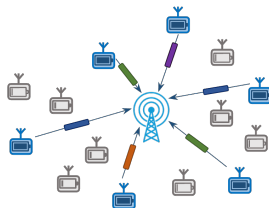
Limitation of UMA

- **Message collisions** treated as errors in UMA
- Message collision: transmission of the same message by multiple users

Do message collisions occur in practice?



Sampling and quantization



Data transmission

- Users may transmit the same quantized information
- Applications: multi-target localization, federated learning, point-cloud transmission
- We are interested in both the set of transmitted messages and their **multiplicities**

Type-based UMA (TUMA)

An extension of UMA that handles message collision

Goal: Estimate type, i.e., empirical distributions of messages

Type estimation idea over multiple access channel (MAC) [Mergen and Tong, 2006]:

- Assumed that each message can be associated to an orthogonal codeword
- However, orthogonalization is impossible (number of possible messages $>$ blocklength)

TUMA over Gaussian MAC [Ngo et al., 2024]

- Type estimation formulated as a compressed sensing problem
- Approximate message passing (AMP) algorithm employed for decoding
- Assumes perfect power control
- Multiplicities estimated from the received codeword power

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Extension to fading is not straightforward

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How can we handle fading channels?

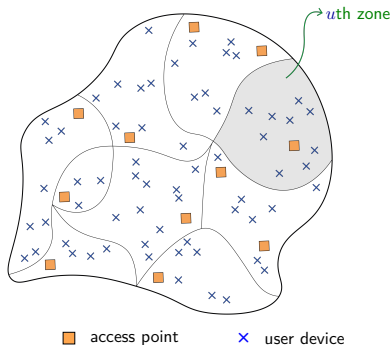
The benefits of distributed MIMO (D-MIMO) architectures in UMA

[Çakmak et al., 2025]: UMA within D-MIMO

- Rayleigh fading channel
- Perfect knowledge of large-scale fading model
- Location-based codebook partition
- Multisource AMP decoder

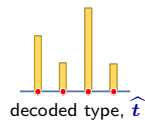
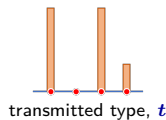
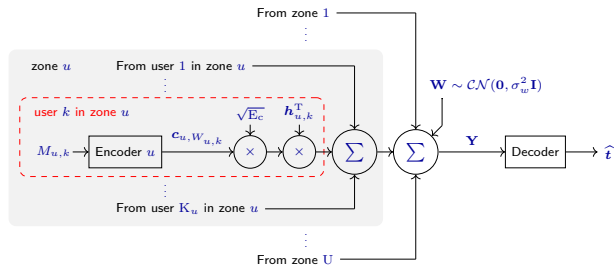
Same ingredients can be used for TUMA under fading

TUMA over a D-MIMO architecture



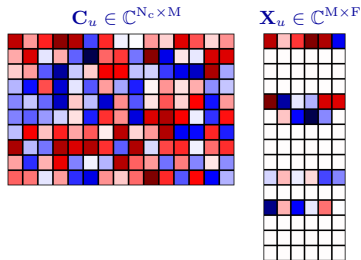
- Area partitioned into U zones
- Users in a zone have similar large-scale fading coefficients (LSFCs)
- Codebook partitioned into U zones
- Codebook for zone u : $\mathbf{C}_u \in \mathbb{C}^{N_c \times M}$
- N_c : blocklength
- M : number of possible messages
- B access points (APs) with A antennas each
- $F = B \times A$ antennas in total
- APs connected to a CPU via fronthaul links

System overview of TUMA over D-MIMO



- User k in zone u has message $M_{u,k}$ and transmits codeword $c_{u,W_{u,k}}$, scaled by $\sqrt{E_c}$
- Fading modeled via channel vector $\mathbf{h}_{u,k} \in \mathbb{C}^F$ (from user to all AP antennas)
- Transmitted messages $\{M_{u,k}\}$ form the type vector $\mathbf{t} = [t_1, \dots, t_M]^T$
- CPU receives superposition signal $\mathbf{Y}$ and estimates the type $\hat{\mathbf{t}}$ from $\mathbf{Y}$ using known codebooks $\{\mathbf{C}_u\}$ and an LSFC model (no instantaneous CSI)
- **Performance metric:** total variation distance $\mathbb{TV}(\mathbf{t}, \hat{\mathbf{t}}) = \sum_{m=1}^M |t_m - \hat{t}_m|/2$

TUMA as a compressed sensing problem



- Users who transmit the same message contribute to the same row of X_u
- The m th row is:

$$x_{u,m} = \sum_{k: W_{u,k}=m} h_{u,k}$$

- Only a few messages are active $\rightarrow X_u$ is row-sparse

Global system equation:

$$Y = \sqrt{E_c} \sum_{u=1}^U C_u X_u + W$$

This is a structured compressed sensing problem (multisource)

Multisource AMP for type estimation

We use **multisource AMP** [Çakmak et al., 2025] to estimate $\{\mathbf{X}_u\}$ and enable type estimation

For $t = 1, 2, \dots$, iterate:

- Effective observation: $\mathbf{R}_u^{(t)} = \mathbf{C}_u^H \mathbf{Z}^{(t-1)} + \sqrt{E_c} \mathbf{X}_u^{(t-1)}$
- Bayesian denoising: $\mathbf{X}_u^{(t)} = \eta_{u,t}(\mathbf{R}_u^{(t)})$
- Onsager correction: $\mathbf{Q}_u^{(t)}$ via Jacobian of $\eta_{u,t}$
- Residual update: $\mathbf{Z}^{(t)} = \mathbf{Y} - \sqrt{E_c} \sum_{u=1}^U \left(\mathbf{C}_u \mathbf{X}_u^{(t)} - \frac{M}{N_c} \mathbf{Z}^{(t-1)} \mathbf{Q}_u^{(t)} \right)$

Asymptotically, the model $\mathbf{Y} = \sqrt{E_c} \sum_{u=1}^U \mathbf{C}_u \mathbf{X}_u + \mathbf{W}$ decouples into M independent submodels:

$$\mathbf{r}_{u,m}^{(t)} = \sqrt{E_c} \mathbf{x}_{u,m}^{(t)} + \mathcal{CN}(\mathbf{0}; \mathbf{T}^{(t)}),$$

where $\mathbf{T}^{(t)}$ is a diagonal covariance matrix computed via state evolution

The Bayesian denoiser $\eta_{u,t}$ is designed for this decoupled model to estimate $\mathbf{x}_{u,m}^{(t)}$

Proposed centralized TUMA decoder

Our contributions in paper [A]:

Bayesian denoiser $\eta_{u,t}$ is tailored to the TUMA system using:

- Prior on multiplicities $k_{u,m}$: **truncated binomial distribution**
- Prior on user positions: **uniform distribution** over the zone

Challenges due to message collisions:

- High-dimensional integrals in denoiser and Onsager correction
- Solved using **Monte Carlo sampling** for scalability

Type estimation procedure:

- Bayesian estimation of type $\hat{t}$ (using posteriors already computed to evaluate $\eta_{u,t}$)

Proposed distributed TUMA decoder

To increase scalability, we design a **distributed decoder** inspired by **distributed AMP** [Bai et al., 2022]

Key idea: Move likelihood computations from CPU to the access points (APs)

At each AP:

- Process local received signal
- Compute local likelihoods using Monte Carlo sampling

At the CPU:

- Fuse them to form global likelihoods
- Perform Bayesian type estimation

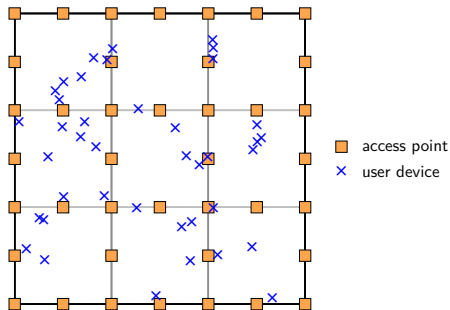
Benefits compared to centralized decoder:

Reduced CPU load, reduced fronthaul bandwidth, scalable to large D-MIMO networks

Drawback:

Slight performance loss compared to centralized decoder

Simulation settings



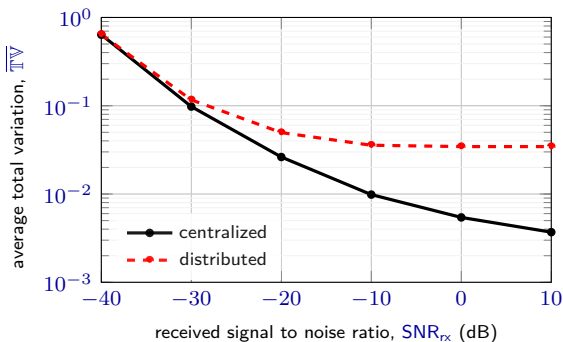
- 300 m $\times$ 300 m coverage area, $U = 9$ zones
- $B = 40$ APs with $A = 4$ antennas each
- Gaussian codebook $\mathbf{C}$, blocklength $N = 1024$
- $M = 2^{12}$ possible messages
 $M > N$: orthogonalization impossible
- 180 active users (20 users per zone),
 avg. multiplicity is 2 and max. multiplicity is 7
- LSFC between user at position $\boldsymbol{\rho}$ and AP b at position $\boldsymbol{\nu}_b$ is modeled as

$$\gamma_b(\boldsymbol{\rho}) = \frac{1}{1 + (\|\boldsymbol{\rho} - \boldsymbol{\nu}_b\|/d_0)^\alpha},$$

where $\alpha = 3.67$ and $d_0 = 13.57$ m

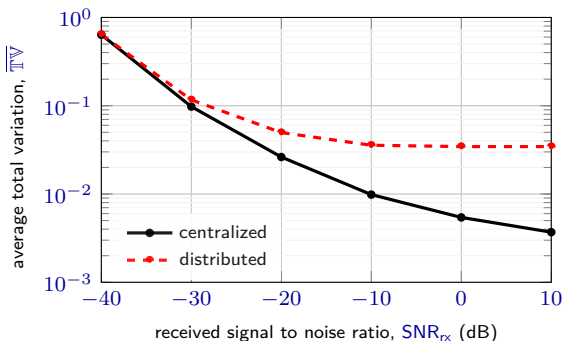
Simulation results

Centralized vs. distributed decoders

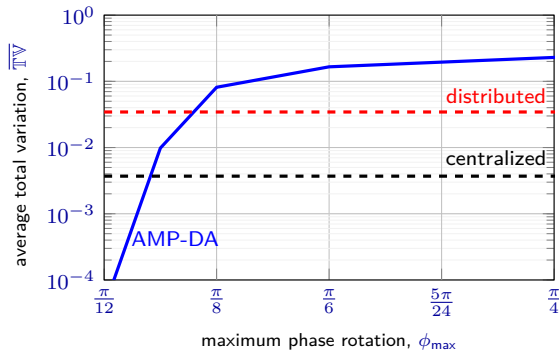


Simulation results

Centralized vs. distributed decoders



Comparison with AMP-DA



- AMP-DA [Qiao et al., 2024]: requires CSI at the users for channel pre-equalization
- Our TUMA decoders: no instantaneous CSI

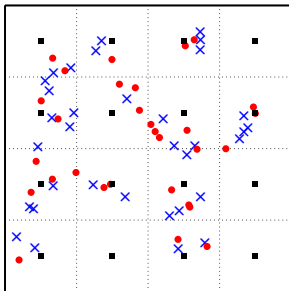
Our decoders outperform AMP-DA even for small phase rotation

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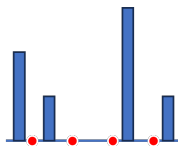
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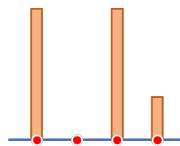
Application I: Multi-target localization



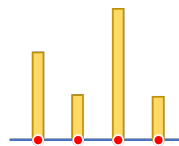
- × sensor
- target
- quantized position



ground truth: ω



quantized type: t



decoded type: $\hat{t}$

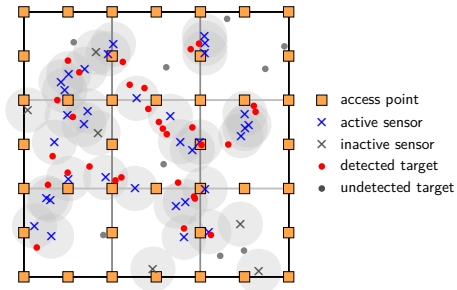
Main aim:

- Sensors transmit target positions to the server via TUMA
- Server estimates the type of quantized positions $\hat{t}$

Some assumptions:

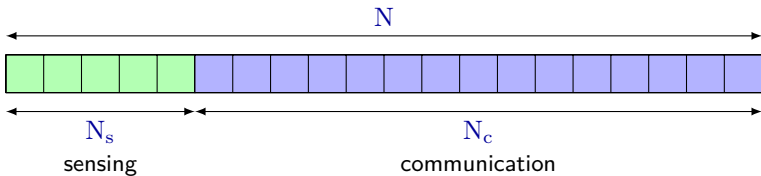
- Snapshot localization
- Each sensor may transmit position of a single target

Application I: Multi-target localization—considered setup



- 200 sensors, 50 targets
- Sensing phase followed by quantization and communication phase: $N = N_c + N_s = 2000$
- Probability of detecting a target at distance d :

$$p_d \propto \frac{N_s P_s}{d^4}$$

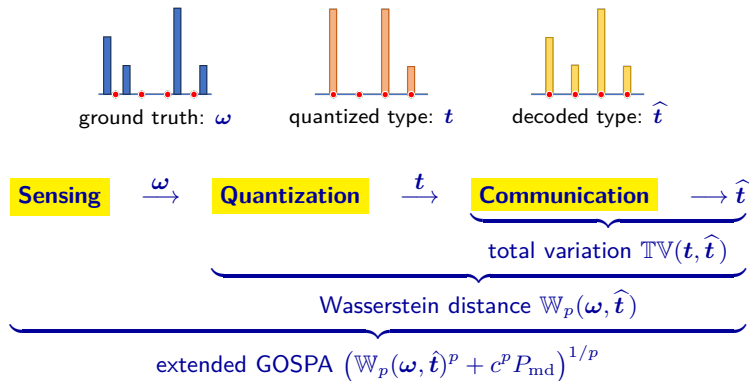


Application I: Multi-target localization–contributions

Our contributions in paper [B]

- Sensing followed by quantization and communication phase model
- Prior on multiplicities $k_{u,m}$ tailored to multi-target localization setup
- Proposed overall performance metrics
- End-to-end tradeoff evaluation

Overall performance metrics



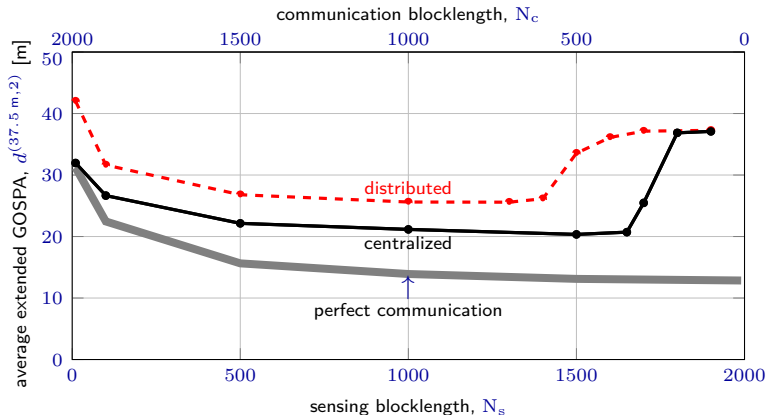
- Quantization and communication performance: $\mathbb{W}_p(\omega, \hat{t})$
- Sensing performance: $P_{\text{md}} = \frac{\# \text{ misdetected targets}}{\# \text{ total targets}}$

Impact of varying N_s and N_c : sensing vs. communication

same D-MIMO network as in paper [A], $\text{SNR}_{\text{rx}} = 10 \text{ dB}$, $P_s = P_c$

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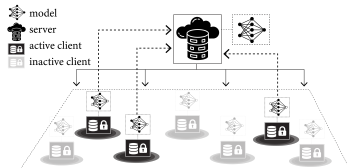
Performance tradeoff between sensing and communication

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Application II: Federated learning



Considered setup:

- Over-the-air federated learning over wireless fading channel
- Client selection under data heterogeneity

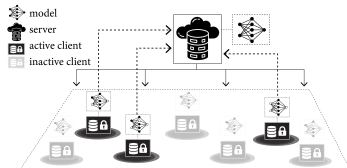
Federated Averaging (FedAvg) [McMahan et al., 2017]

- Clients collaboratively train a shared global model
- Local updates $\Delta^{(t)} \in \mathbb{R}^W$ are transmitted over a shared wireless medium and aggregated at the server

Key question: How can we design a **client selection** + **unsourced transmission scheme** that:

1. preserves client privacy,
2. remains reliable under fading and interference?

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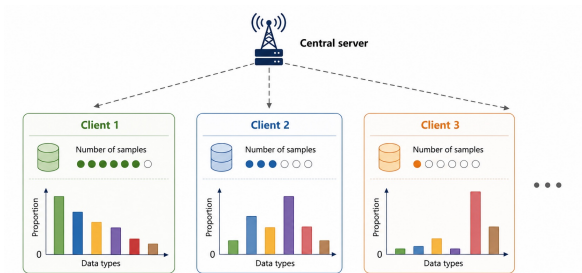
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Our contributions in paper [C]

- Client self-selection strategy, no server-side client information required
- TUMA-based model aggregation over D-MIMO fading channels using [A]

Data heterogeneity and client selection



- Only a **subset of clients** can participate in each communication round
- **Random client selection performs poorly** under heterogeneous local datasets
- **Existing** methods rely on **server-side client information**, violating privacy
- Power-of-choice (PoC) [Choi et al., 2022]: clients with **larger local training losses** are preferred
- PoC: clients **report local losses** to the server for selection, partially violating privacy

Our proposed client self-selection strategy

Key ideas:

- Inspired by PoC [Cho et al., 2022]: clients with higher local loss participate with higher probability
- Unlike PoC, self-selection is performed locally at clients
- The server adjusts a global threshold to control the number of participating clients

Self-selection procedure (per round t): target number of selected clients K_{tar}

1. **Candidate set:** Active clients enter the candidate set randomly
2. **Self-selection:** Candidate client k computes its loss $\ell_k^{(t)}$ and participates with probability

$$\text{sigmoid}\left(a\left(\ell_k^{(t)} - \theta^{(t)}\right)\right)$$

3. **Threshold update:** The server adjusts the threshold as

$$\theta^{(t+1)} \leftarrow \theta^{(t)} + \xi\left(\widehat{L}^{(t)} - K_{\text{tar}}\right),$$

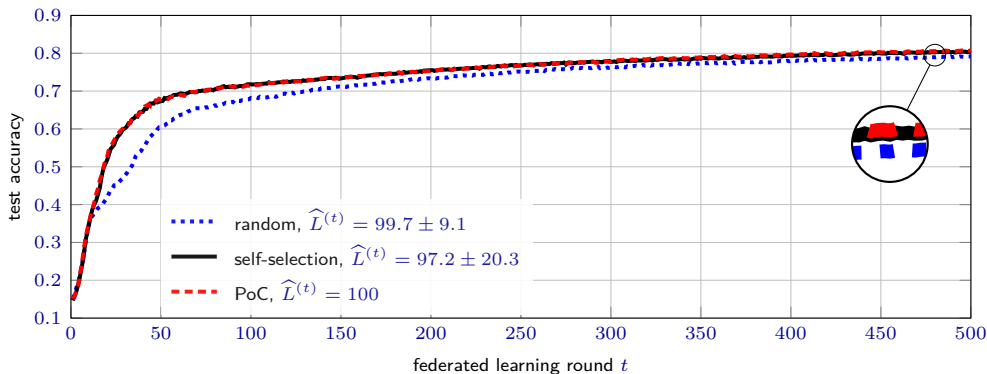
where $\widehat{L}^{(t)}$ is the estimated number of participating clients

Simulation results under perfect communication

image classification over FMNIST dataset, moderate data heterogeneity,
total number of clients $K = 1000$, target number of selected clients $K_{\text{tar}} = 100$

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The proposed self-selection works close to the PoC and outperforms random selection

Simulation results over D-MIMO networks

same D-MIMO network as in paper [A] and [B],

image classification over FMNIST, self-selection, moderate data heterogeneity,
total number of clients $K = 1000$, target number of selected clients $K_{\text{tar}} = 100$,

vector dimension $Q = 30$, number of bits $J = 7$ ($2^J > N_c$: orthogonalization impossible)

benchmark: MD-AirComp [Qiao et al., 2024] scheme that uses AMP-DA

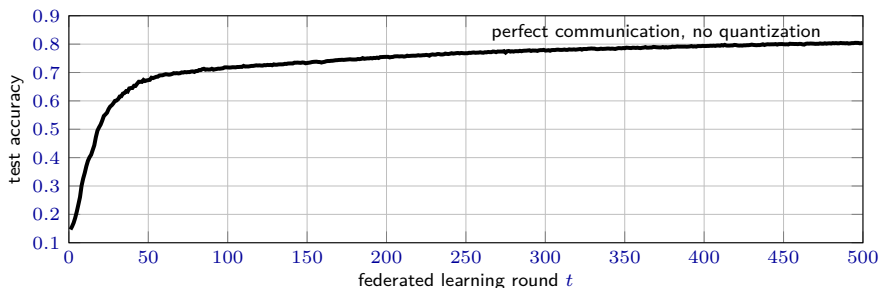
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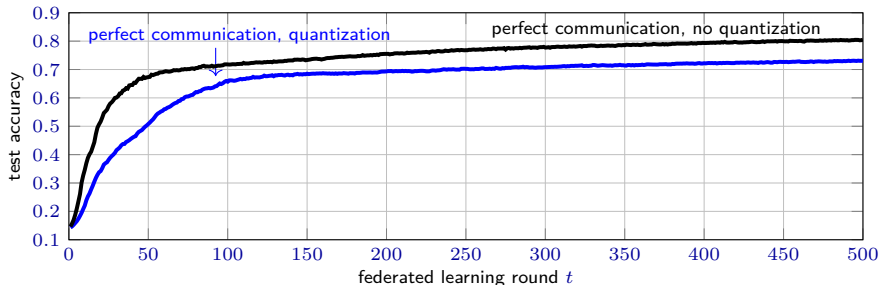
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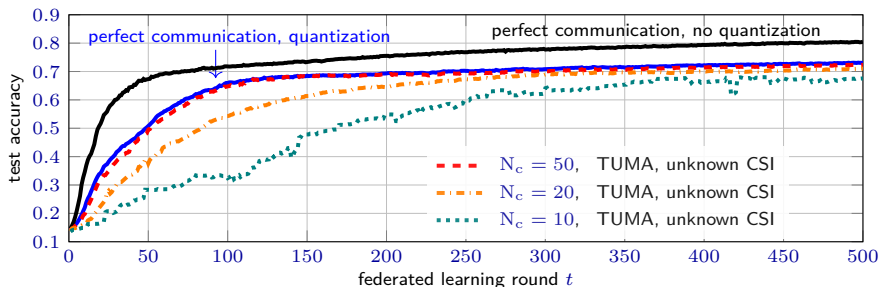
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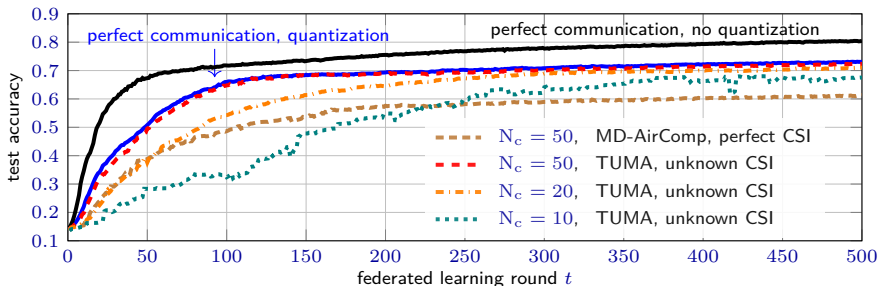
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Our proposed method is close to perfect communication with quantization even without CSI

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- We extended the **TUMA** framework to **fading channels** using a **D-MIMO** architecture
- Proposed **centralized** and **distributed decoders** based on multisource AMP
- Simulation results show improved **type estimation** and **robustness compared to AMP-DA**
- We studied two applications:

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 - **End-to-end** evaluation of sensing and communication tradeoffs

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 - Performance **close to PoC** under **data heterogeneity**
 - Improved performance over **MD-AirComp** even without requiring **CSI**

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